

Experiment:5 implement the Vacuum Cleaner Problem using Python

Aim

o implement the Vacuum Cleaner Problem using Python.

Algorithm

1. Assume two rooms: A and B.
2. Check the current location of the vacuum cleaner.
3. If the room is dirty, clean it.
4. Move to the next room.
5. Repeat until all rooms are clean.
6. Display the final state of the rooms.

Program

```
# Vacuum Cleaner Problem
```

```
rooms = {'A': 'Dirty', 'B': 'Dirty'}
```

```
location = 'A'
```

```
while True:
```

```
    print("Vacuum is in Room", location)
```

```
    if rooms[location] == 'Dirty':
```

```
        print("Cleaning Room", location)
```

```
        rooms[location] = 'Clean'
```

```
    if location == 'A':
```

```
        location = 'B'
```

```
    else:
```

```
        break
```

```
print("\nFinal State:")
```

```
for room in rooms:
```

```
    print(room, ":", rooms[room])
```

```
exp 3 water jug problem.py - C:/Users/yadal/OneDrive/Documents/AI/exp 3 water jug probl...
File Edit Format Run Options Window Help
# Water Jug Problem
jug1 = 4
jug2 = 3
target = 2

print("Jug1\tJug2")

a = 0
b = 0

while a != target and b != target:
    if a == 0:
        a = jug1
        print(a, "\t", b)
    elif b == jug2:
        b = 0
        print(a, "\t", b)
    else:
        transfer = min(a, jug2 - b)
        a -= transfer
        b += transfer
        print(a, "\t", b)

print("\nTarget reached!")
```

Sample Output

```
>>> ===== RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp 5 vacume cleaner.py =====
Vacuum is in Room A
Cleaning Room A
Vacuum is in Room B
Cleaning Room B

Final State:
A : Clean
B : Clean
\\>>>
```

Result

The Vacuum Cleaner Problem was successfully implemented in Python. The vacuum cleaner cleaned all dirty rooms and reached the goal state where both rooms were clean.